

$$f_{ij}^* : i \in C.$$

$$f_{ij}^* = \begin{cases} 0 & \text{if } i \notin C, i \in C, j \in D, C \neq D \\ 1 & \text{if } i \in C, i, j \in C \\ \alpha_i^{(C)} & \text{if } i \text{ tran}, i \in C, C \text{ rec.} \\ j & \text{if } i, j \text{ tran}, i \neq j \end{cases}$$

$$\tilde{P} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

$$\tilde{C} = \{i\}, \quad \tilde{P}, \quad \alpha_i^{(\tilde{C})}$$

$$\begin{aligned}
f_{ii}^* &= f_{ii}^{(1)} + \sum_{n=2}^{\infty} f_{ii}^{(n)} \\
&= p_{ii} + \sum_{n=2}^{\infty} \sum_{j \in T} p_{ij} f_{ji}^{(n-1)} \\
&= p_{ii} + \sum_{j \in T} p_{ij} \sum_{n=1}^{\infty} f_{ji}^{(n)} \quad \leftarrow f_{ii}^* \\
&= p_{ii} + \sum_{j \in T} p_{ij} f_{ji}^*
\end{aligned}$$

Stationary distⁿ.

$$\pi(a) = \frac{\gamma(a)}{R} = \sum_{j \in S} p_{ja} \frac{\gamma(j)}{R}$$

"thermodynamic
equilibrium"
Stationarity.



$$\gamma(a) \cdot \frac{1}{2}$$

$a \rightarrow b$

$$\gamma(b) \cdot \frac{1}{3}$$

$$R = \sum \gamma(i)$$

We call a prob. distrⁿ π on the state space a stationary distrⁿ if

$$\pi = \pi P. \quad \text{---} \quad (*)$$

but $\sum m(i) = \infty$

$$\pi = (\pi(i) : i \in S), \quad \sum_{i \in S} \pi(i) = 1.$$

$$\pi(i) \geq 0.$$

In some cases, $\exists m = (m(i) : i \in S)$ s.t.
 $m(i) \geq 0, \quad m = mP.$

The eqⁿ (*) is called the balance eqⁿ.

$$\mathcal{I}(i) = \sum_{j \in S} \pi(j) p_{ji} \quad \forall i \in S.$$

If $\pi(i) p_{ij} = \pi(j) p_{ji}$ for all $i, j \in S$,
then we say the system is reversible in π .

and in physics notation, it is called detailed balance.

$\frac{\text{Prop}^n}{\text{in } \mathbb{T}}$ If the chain is reversible then π is a st. distⁿ

Pf $\nearrow$:-

For any $i \in S$,

$$\pi(i) = \sum_{j \in S} \pi(i) p_{ij} = \sum_{j \in S} \pi(j) p_{ji}$$

Ex:- $S = \{0, 1, 2\}$. $\pi = (\lambda, 0, 1-\lambda)$
 $\Rightarrow \pi P = \pi$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1/3 & 1/6 \\ 0 & 0 & 1 \end{bmatrix} \quad \pi_1 = (1, 0, 0)$$

$$\pi_1 = \pi_1 P$$

Also, $\pi_2 = (0, 0, 1)$, $\pi_2 = \pi_2 P$.

Note, $\lambda \pi_1 + (1-\lambda) \pi_2$ for any $\lambda \in [0, 1]$
 is a st. distribⁿ

Ex:- $S = \{1, 2\}$,

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \quad 0 < p_{12}, p_{21} < 1.$$

$$\pi = \left(\frac{p_{21}}{p_{12} + p_{21}}, \frac{p_{12}}{p_{12} + p_{21}} \right)$$

$$\pi P = \pi.$$

Ex:

$$P = \begin{array}{c|cccc} & 0 & 1 & 2 & 3 \\ \hline 0 & 0 & 1 & & \\ 1 & 0 & 0 & 1 & \\ 2 & 0 & 0 & 0 & 1 \end{array}$$

$$p_{i,i+1} = 1$$

$\mathbb{I}$

$$\exists \pi \text{ s.t.} \\ \pi = \pi P.$$

In generic example, for
the classes $C_1 = \{0, 3\}$, $C_2 = \{2, 4, 7\}$.

Let μ be any prob on S .
" $(\mu(i) : i \in S)$.
 $\sum \mu(i) = 1$,
 $\mu(i) \geq 0$

π_1 π_2

$X_0 \sim \mu$, i.e., $P(X_0 = i) = \mu(i)$.

$$P(X_1=i) = \sum_{j \in S} P(X_1=i | X_0=j) \underline{P(X_0=j)}.$$

$$= \sum_{j \in S} \mu(j) p_{ji}$$

$$= (\mu P)_i$$

In other words,

$$X_1 \sim \mu P.$$

By induction,

$$X_n \sim \mu P^n$$

$$P(X_n=i)$$

$$= \sum_{j \in S} P(X_n=i | X_0=j) \mu(j)$$

$$= \sum_j \mu(j) p_{ji}^{(n)}$$

$$= \sum_{j \in S} \mu(j) p_{ji}^n$$

$$= (\mu P^n)(i)$$

If $\mu = \pi$, π st. distrⁿ.

$$X_n \sim \pi \quad \forall \underline{n \geq 0}.$$

Thm:- For a irr. pos. rec.

chain, there exists a unique st. distrⁿ.

Cor:- For every pos. rec class C , there exists a unique st. distrⁿ π_C which gives

mass only to elements of C .

Thm:- For an irr. pos. rec. chain,

$$\pi(i) = \frac{1}{\mu_i}$$

, where $\mu_i = E(T_i | X_0 = i)$

Ex:- If for a finite ^{irr.} chain, the tr. matrix is doubly stochastic, then the st. distⁿ is $\frac{1}{N}(1, \dots, 1)$.

T_i is the first return time to i .

Note, for $\underline{1} = (1, \dots, 1)$,

$$\underline{1}^T P = \underline{1}^T.$$

$$\Rightarrow \frac{1}{N} \underline{1}^T P = \frac{1}{N} \underline{1}^T.$$

Now, $\frac{1}{N} \underline{1}^T$ is a prob. & hence a st. distrib.
Since for pos. val., the st. distrib. is unique,
 $\frac{1}{N} \underline{1}^T$ is the only st. distrib.

Sp. a dice is being rolled. Let ϵ_i
the score of i th roll.

Define, $X_n = \left(\sum_1^n \epsilon_i \right) \bmod (N)$

$$P(X_n = i) \rightarrow \frac{1}{N} \quad \begin{matrix} 0 & 1 & 2 & \dots & N-1 \end{matrix}$$

P is a doubly st.
matrix.

X_n is irr.

$$P = \begin{pmatrix} 0 & 1 & 2 & \dots & N-1 \\ 1 & 2 & 3 & \dots & 0 \\ 2 & 3 & 4 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix}$$

Hence, $\frac{1}{N} \mathbf{1}$ is the only st. distrⁿ.

i.e., $\frac{1}{\mu_i} = \pi(i) = \frac{1}{N} = \lim_{n \rightarrow \infty} p_{ii}^{(n)} \geq \frac{1}{N}$.

$\lim_{n \rightarrow \infty} p_{ii}^{(n)}$ = $\begin{cases} 0 & \text{if } i \text{ rec, or tran} \\ \frac{1}{\mu_i} & \text{if pos. rec, aperiodic} \\ \text{does not exist} & \text{pos. rec, period} \geq 1. \end{cases}$

$$\lim_{n \rightarrow \infty} P(X_n = i) = \lim_{n \rightarrow \infty} \sum_{j \in S} P(X_n = i | X_0 = j) P(X_0 = j)$$

$$= \sum_{j \in S} \left[\lim_{n \rightarrow \infty} p_{ji}^{(n)} \right] P(X_0 = j)$$

$$= \frac{1}{\mu_i} \sum_{j \in S} P(X_0 = j) \mu_j$$

$$= \frac{1}{\mu_i}$$

RW on Graphs.

Ass. that the RW is
irr. Finite st. sp.

Hence, has a
unique st. distrⁿ.

$$\pi(i) = \frac{w_{i.}}{w_{..}}, \quad i \in S.$$

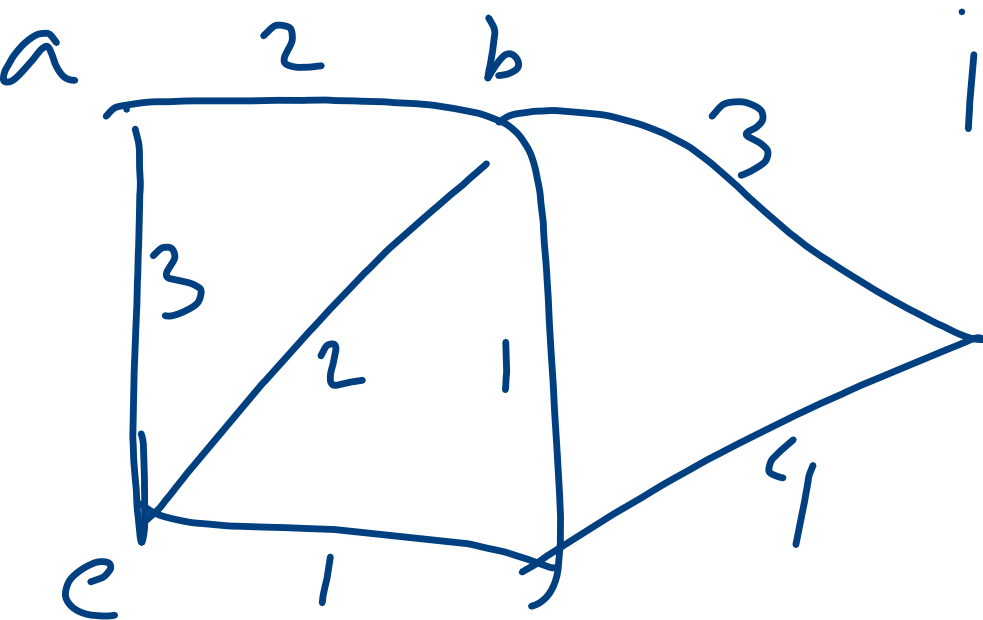
$$w_{ij} = w_{ji}$$

$$w_{i.} = \sum_{j \in S} w_{ij} > 0$$

$$w_{..} = \sum_{i \in S} w_{i.}$$

$$p_{ij} = \frac{w_{ij}}{w_{i.}}$$

$$\sum_{i \in S} \pi(i) = 1$$



$$\pi(i) P_{ij} = \frac{w_{i.}}{w_{..}} \cdot \frac{w_{ij}}{w_{i.}} = \frac{w_{ij}}{w_{..}}$$

$$\pi(j) P_{ji} = \frac{w_{j.}}{w_{..}} \cdot \frac{w_{ji}}{w_{j.}} = \frac{w_{ji}}{w_{..}} = \frac{w_{ij}}{w_{..}}$$

Look on Left Bottom Sq. How much time does it take to return?